We'll see now how Schur's lemma allows us to classify subrepresentations in finite dimensional semisimple representations.

Proposition 3.1.4. *Let $V_i, 1 \le i \le m$, be irreducible finite dimensional pairwise nonisomorphic representations of A, and let W be a subrepresentation of $V = \bigoplus_{i=1}^m n_i V_i$. Then W is isomorphic to $\bigoplus_{i=1}^m r_i V_i$, $r_i \le n_i$, and the inclusion $\phi : W \to V$ is a direct sum of inclusions $\phi_i : r_i V_i \to n_i V_i$ given by multiplication of a row vector of elements of V_i (of length r_i) by a certain $r_i \times n_i$ matrix X_i with linearly independent rows: $\phi(v_1, \dots, v_{r_i}) = (v_1, \dots, v_{r_i}) X_i$.*

Proof. The proof is by induction in $n := \sum_{i=1}^m n_i$. The base of induction ($n = 1$) is clear. To perform the induction step, let us assume that W is nonzero, and fix an irreducible subrepresentation $P \subset W$. Such P exists (Problem 2.3.15).[1] Now, by Schur's lemma, P is isomorphic to V_i for some i, and the inclusion $\phi|_P : P \to V$ factors through $n_i V_i$ and upon identification of P with V_i is given by the formula $v \mapsto (vq_1, \dots, vq_{n_i})$, where $q_l \in k$ are not all zero.

Now note that the group $G_i = GL_{n_i}(k)$ of invertible $n_i \times n_i$ matrices over k acts on $n_i V_i$ by $(v_1, \dots, v_{n_i}) \mapsto (v_1, \dots, v_{n_i}) g_i$ (and by the identity on $n_j V_j$, $j \ne i$) and therefore acts on the set of subrepresentations of V, preserving the property we need to establish: namely, under the action of g_i, the matrix X_i goes to $X_i g_i$, while the matrices $X_j, j \ne i$, don't change. Take $g_i \in G_i$ such that $(q_1, \dots, q_{n_i}) g_i = (1, 0, \dots, 0)$. Then $W g_i$ contains the first summand V_i of $n_i V_i$ (namely, it is $P g_i$); hence $W g_i = V_i \oplus W'$, where $W' \subset n_1 V_1 \oplus \dots \oplus (n_i - 1) V_i \oplus \dots \oplus n_m V_m$ is the kernel of the projection of $W g_i$ to the first summand V_i along the other summands. Thus the required statement follows from the induction assumption. $\square$

Remark 3.1.5. In Proposition 3.1.4, it is not important that k is algebraically closed, nor does it matter that V is finite dimensional. If these assumptions are dropped, the only change needed is that the entries of the matrix X_i are no longer in k but in $D_i = \mathrm{End}_A(V_i)$,

[1] Another proof of the existence of P, which does not use the finite dimensionality of V, is by induction in n. Namely, if W itself is not irreducible, let K be the kernel of the projection of W to the first summand V_1. Then K is a subrepresentation of $(n_1 - 1)V_1 \oplus \dots \oplus n_m V_m$, which is nonzero since W is not irreducible, so K contains an irreducible subrepresentation by the induction assumption.